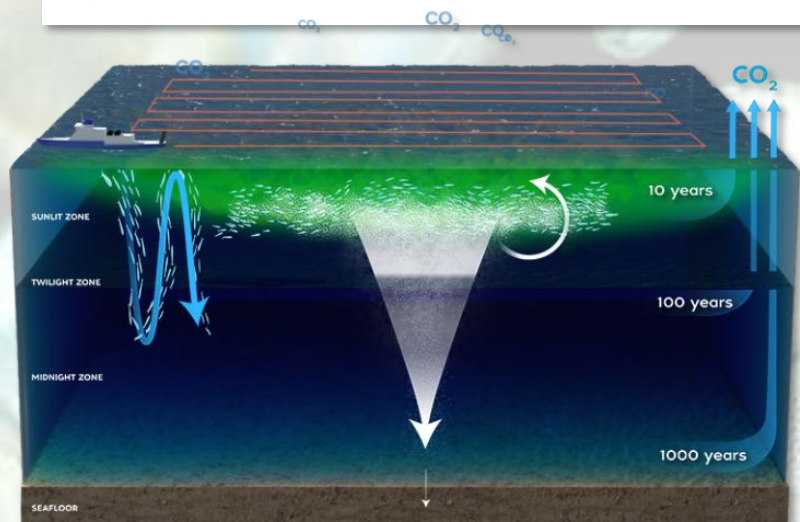


Does Ocean Iron Fertilisation Work as a Carbon Sequestration Method?



Introduction:

Ocean iron fertilisation has a contentious history as a possible method of carbon dioxide removal and storage.

Iron is a limiting micronutrient for phytoplankton, meaning that it limits growth and that a small amount of iron produces a large amount of growth. Phytoplankton absorb CO₂ at the ocean's surface, and when they die sink to lower levels.

The theory behind iron fertilisation is that adding iron to the ocean will produce a phytoplankton 'bloom' – boosted growth, which will result in greater absorption and sequestration.

Image source: Woods Hole Oceanographic Institution

To be considered 'permanently' sequestered, carbon must remain out of the atmosphere for at least 100 years. To achieve this, absorbed carbon must sink below the range of wind mixing (~100m).

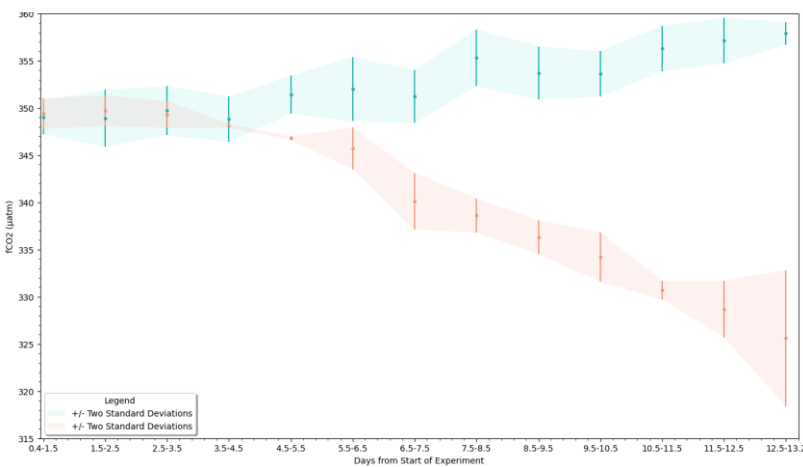
Complications:

- Often phytoplankton are consumed close to the surface, meaning that the CO₂ is released back into the atmosphere
- Physical mixing of the ocean can prevent carbon from sinking
- Diatoms (relatively heavy, so more likely to sink to ocean floor) become lighter when not iron deficient

SOIREE and SOFeX:

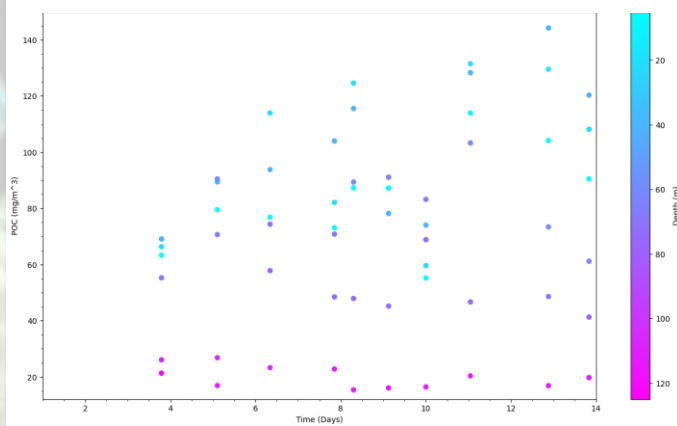
- February 1999, January-February 2002 respectively
- Southern Ocean (high-nutrient, low-chlorophyll region)

fCO₂ Outside and at Centre of Patch (SOIREE)



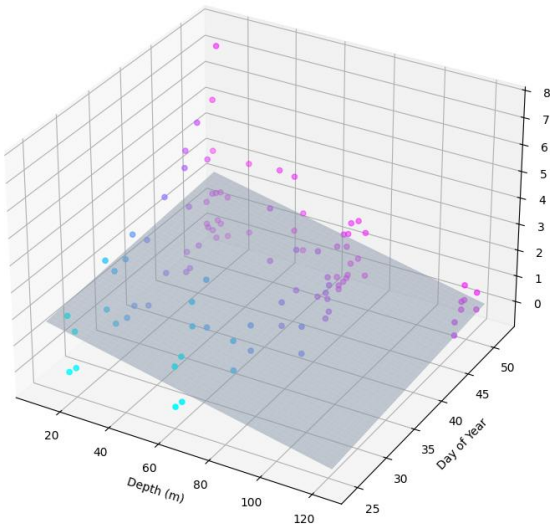
fCO₂ (corrected partial pressure of CO₂) outside and at the centre of the phytoplankton bloom. fCO₂ decreases at the centre, indicating increased drawdown prompted by fertilisation.

Particulate Organic Carbon (SOIREE)

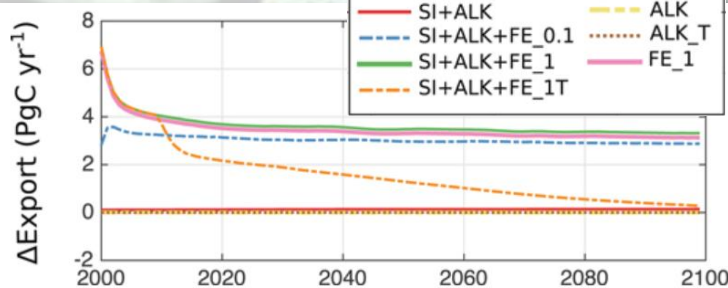


POC over the duration of SOIREE, coloured according to depth. The amount of carbon measured decreases steadily with depth.

POC vs. Day of Year and Depth (SOFeX)



Three-dimensional model fit to variation in measured POC with time and depth. Both factors correspond to a roughly linear decrease in POC.



Hauck et al. uses the marine regulated ecosystem and biochemistry model REcoM2 and the MIT general circulation model (MITgcm) to predict carbon export over a 100-year period under various conditions. The pink line represents the influence of iron on export, with a standard bioavailability of 1%.

Findings:

Ocean iron fertilisation is effective at producing phytoplankton blooms and boosting carbon absorption at the surface; however, results indicate that it is not able to export a sufficient amount of this carbon below the range of mixing to achieve significant permanent sequestration.